

QUICK REFERENCE

Cloud Computing Essentials

A one-sitting reference for core concepts, pricing, and security

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Companion materials

- Slide deck: Cloud_Computing_Session_Nisha.pptx (session walkthrough)
- Handout: Cloud_Computing_Handout_Nisha.docx (full notes + lab steps)
- This PDF: fast lookup during and after the session

1. What Is Cloud Computing?

On-demand access to shared computing resources — servers, storage, databases, networking, software — delivered over the internet, provisioned in minutes, billed for what you use. **Rent, don't buy.**

The 5 NIST characteristics

- **On-demand self-service** — provision without a human in the loop
- **Broad network access** — reachable from any device, anywhere
- **Resource pooling** — multi-tenant infrastructure shared securely
- **Rapid elasticity** — scale out and in automatically with demand
- **Measured service** — metered usage, pay-as-you-go billing

2. Service Models

Model	You manage	Examples
IaaS	OS, runtime, app, data	EC2, Azure VMs, Compute Engine
PaaS	App code and data only	Elastic Beanstalk, App Service, App Engine
SaaS	Users and settings only	Gmail, Microsoft 365, Salesforce

Serverless (FaaS — AWS Lambda, Azure Functions) sits between PaaS and SaaS.

3. Deployment Models

Model	Description
Public	Shared provider infrastructure — fastest start, no hardware
Private	Dedicated to one org — more control, higher cost
Hybrid	Private + public connected — keep sensitive data local
Multi-cloud	2+ public providers — avoids lock-in, adds complexity

4. Core Building Blocks Across Providers

Category	AWS	Azure / GCP
Virtual machines	EC2	Azure VMs / Compute Engine
Object storage	S3	Blob Storage / Cloud Storage
Managed SQL	RDS / Aurora	Azure SQL DB / Cloud SQL
NoSQL	DynamoDB	Cosmos DB / Firestore
Serverless functions	Lambda	Azure Functions / Cloud Run
Managed Kubernetes	EKS	AKS / GKE
Identity & access	IAM	Entra ID / Cloud IAM

5. Shared Responsibility & Security Checklist

You own security IN the cloud

- Data classification & encryption
- Identity & access (IAM), MFA
- App code & OS patching (IaaS)
- Network configuration

Day-one checklist

- 1 Enable MFA on root, then stop using it daily
- 2 Least-privilege IAM roles; review often
- 3 Encrypt at rest (KMS) and in transit (TLS)
- 4 No secrets in code — use a secrets manager
- 5 Turn on audit logging + anomaly alerts
- 6 Back up regularly and test restores

6. Pricing Models

Model	Commitment	Best for
On-Demand	None	Spiky, unpredictable workloads
Savings Plans / RI	1–3 years	Steady baseline load
Spot	None (reclaimable)	Batch jobs, CI runners

Max discounts vs. On-Demand (AWS, published): Savings Plans / RIs up to 72%; Spot up to 90%. Set a billing alarm before you provision anything.

7. Lab: Host a Static Website (~20 min)

- 1 Create an S3 bucket with a globally unique name
- 2 Upload index.html and error.html
- 3 Enable static website hosting on the bucket
- 4 Add a CloudFront distribution with Origin Access Control
- 5 Open the CloudFront URL and confirm the page loads
- 6 Clean up: delete the distribution and the bucket

```
aws s3 mb s3://nisha-site-2026
aws s3 sync ./site s3://nisha-site-2026
aws s3 website s3://nisha-site-2026 --index-document index.html
```

8. Cloud-Native Pipeline

Code (Git/PR) → **Build & test** (CI) → **Containerise** (Docker + scan) → **Deploy** (Kubernetes, GitOps) → **Observe** (metrics, logs, traces)

DevSecOps: scan code (SonarQube), dependencies and images (Trivy, OWASP) inside the pipeline, before production.

9. Suggested Learning Path

Timeframe	Focus	Details
Months 0–2	Foundations	Cloud Practitioner / AZ-900; Linux, networking, Git
Months 2–5	Associate	Solutions Architect Associate; 3 projects with IaC
Months 5–9	Specialise	Kubernetes (CKA), DevOps, or Cloud AI

Key Takeaways

Remember these five points

- Cloud = on-demand, elastic, metered resources
- Pick the service model that minimises what you must manage
- You own security IN the cloud: IAM, data, configuration
- Match pricing to workload shape and set billing alarms
- Automate everything: IaC, CI/CD, observability